Practical 8

Data Visualization I

[ ]:
```
## Data Visualization I
```

[9]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

dataset = sns.load_dataset('titanic')
dataset.head()
```

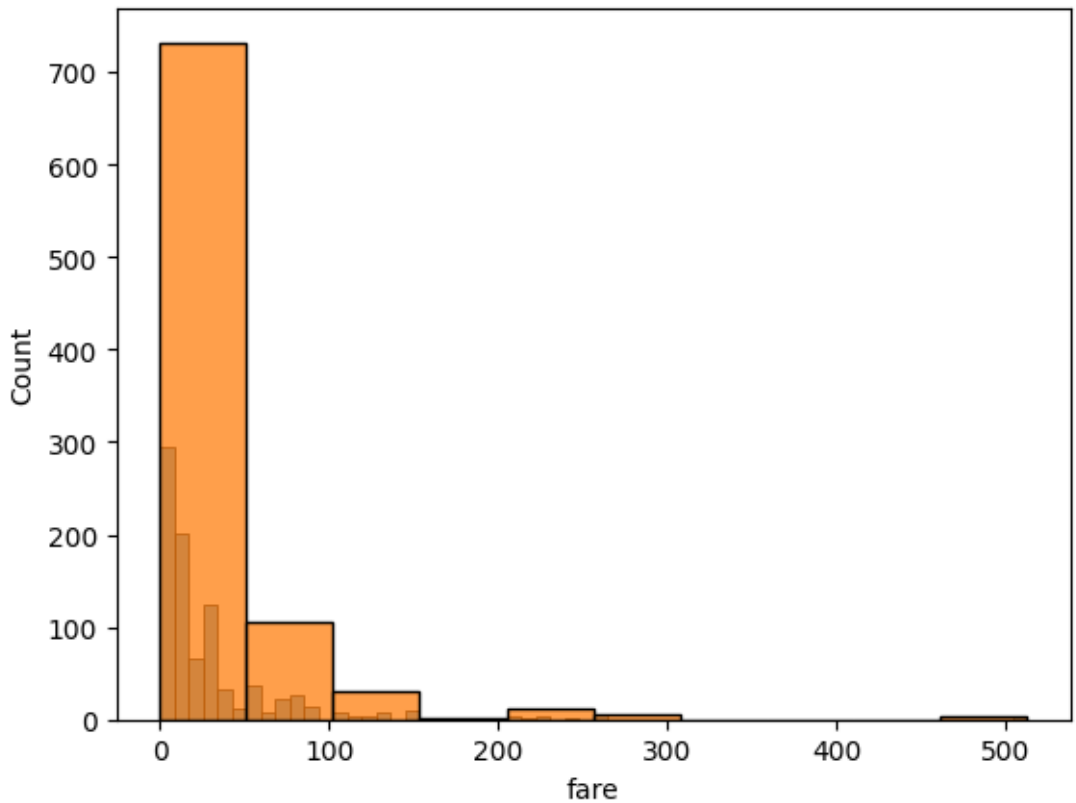
[9]:
```
   survived  pclass     sex   age  sibsp  parch     fare embarked  class  \
0         0       3    male  22.0      1      0   7.2500        S  Third
1         1       1  female  38.0      1      0  71.2833        C  First
2         1       3  female  26.0      0      0   7.9250        S  Third
3         1       1  female  35.0      1      0  53.1000        S  First
4         0       3    male  35.0      0      0   8.0500        S  Third

     who  adult_male deck  embark_town alive  alone
0    man        True  NaN  Southampton    no  False
1  woman       False    C    Cherbourg   yes  False
2  woman       False  NaN  Southampton   yes   True
3  woman       False    C  Southampton   yes  False
4    man        True  NaN  Southampton    no   True
```

[ ]:
```
## Distribution Plot of Ticket Prices (Fare)
```

[23]:
```
sns.histplot(dataset['fare'])
sns.histplot(dataset['fare'], kde=False, bins=10)
```

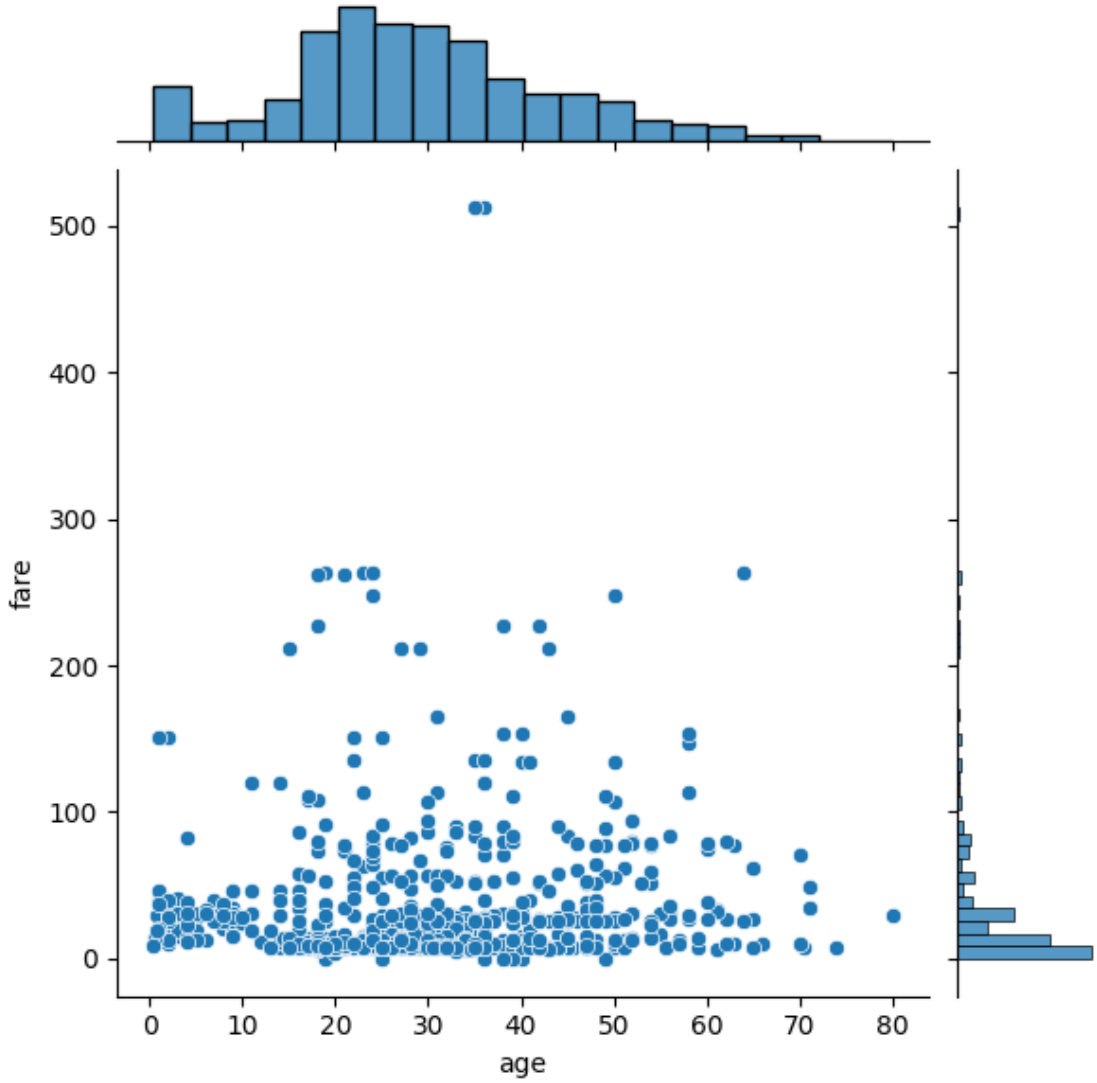
[23]:
```
<Axes: xlabel='fare', ylabel='Count'>
```

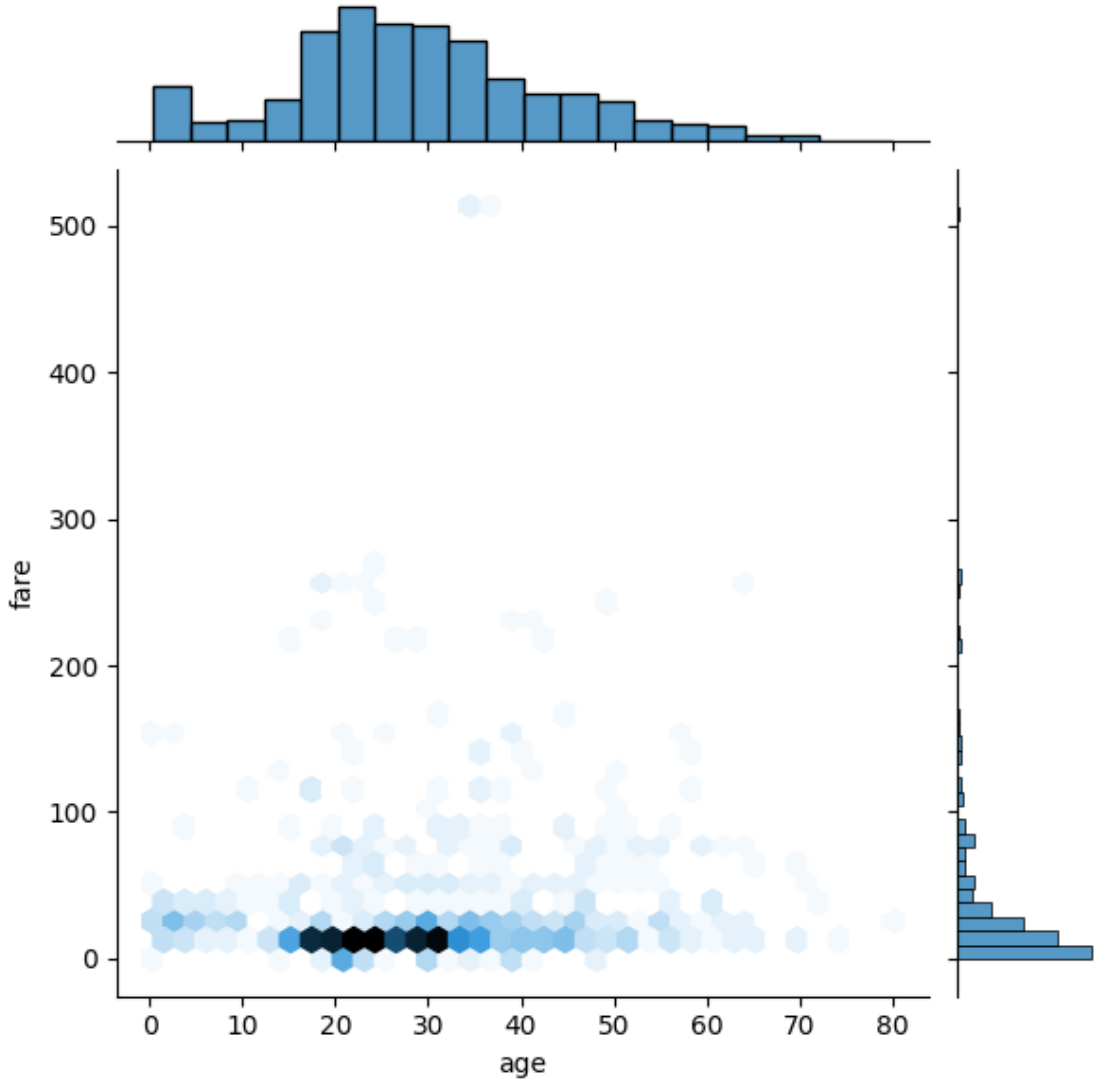


[20]:
```
## Joint Plot of Age vs Fare
```

[24]:
```
sns.jointplot(x='age', y='fare', data=dataset)
sns.jointplot(x='age', y='fare', data=dataset, kind='hex')
```

[24]: <seaborn.axisgrid.JointGrid at 0x708b7ba3c650>

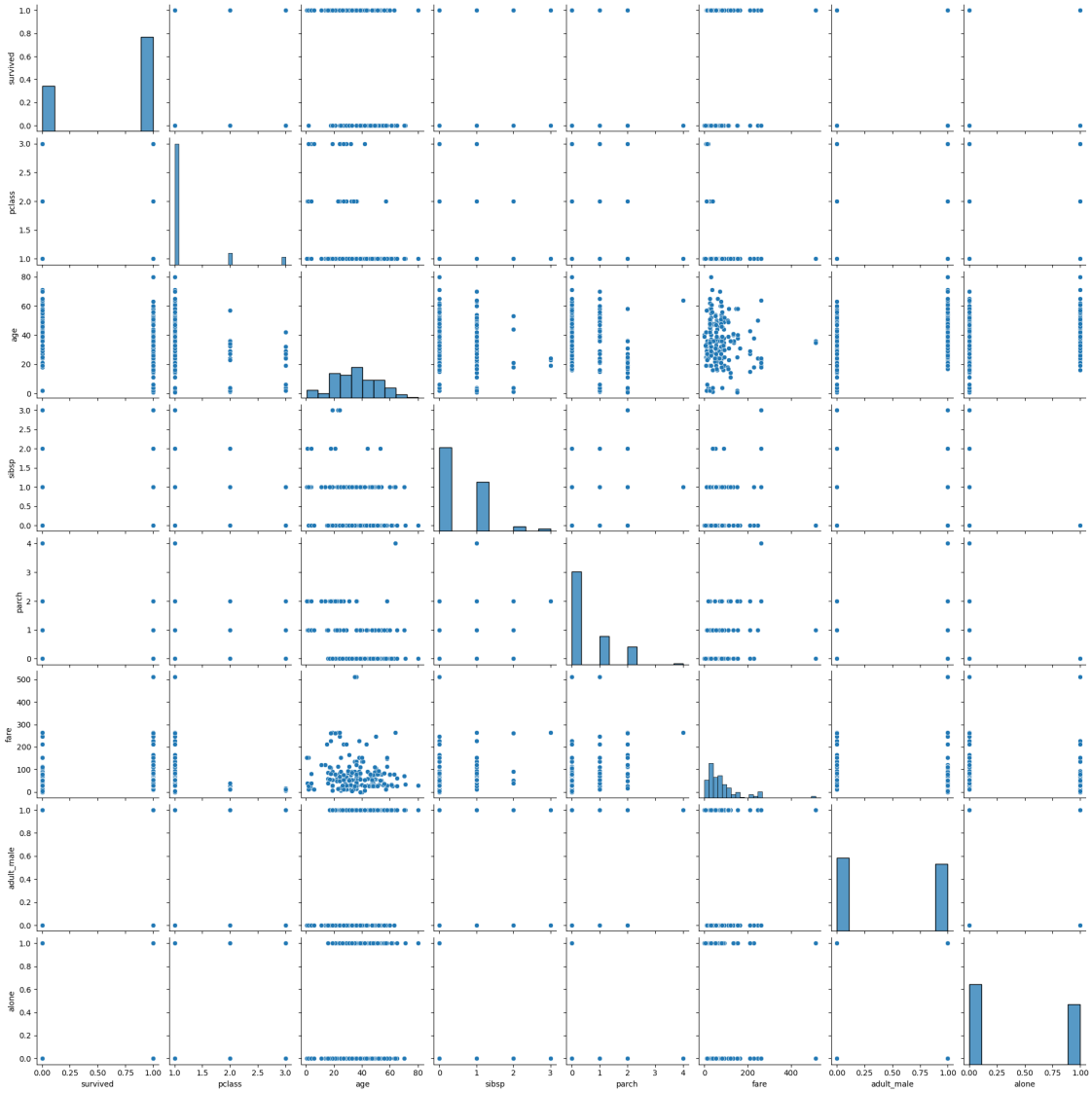

500
400
300
200
100
0
fare
0
10
20
30
40
50
60
70
80
age

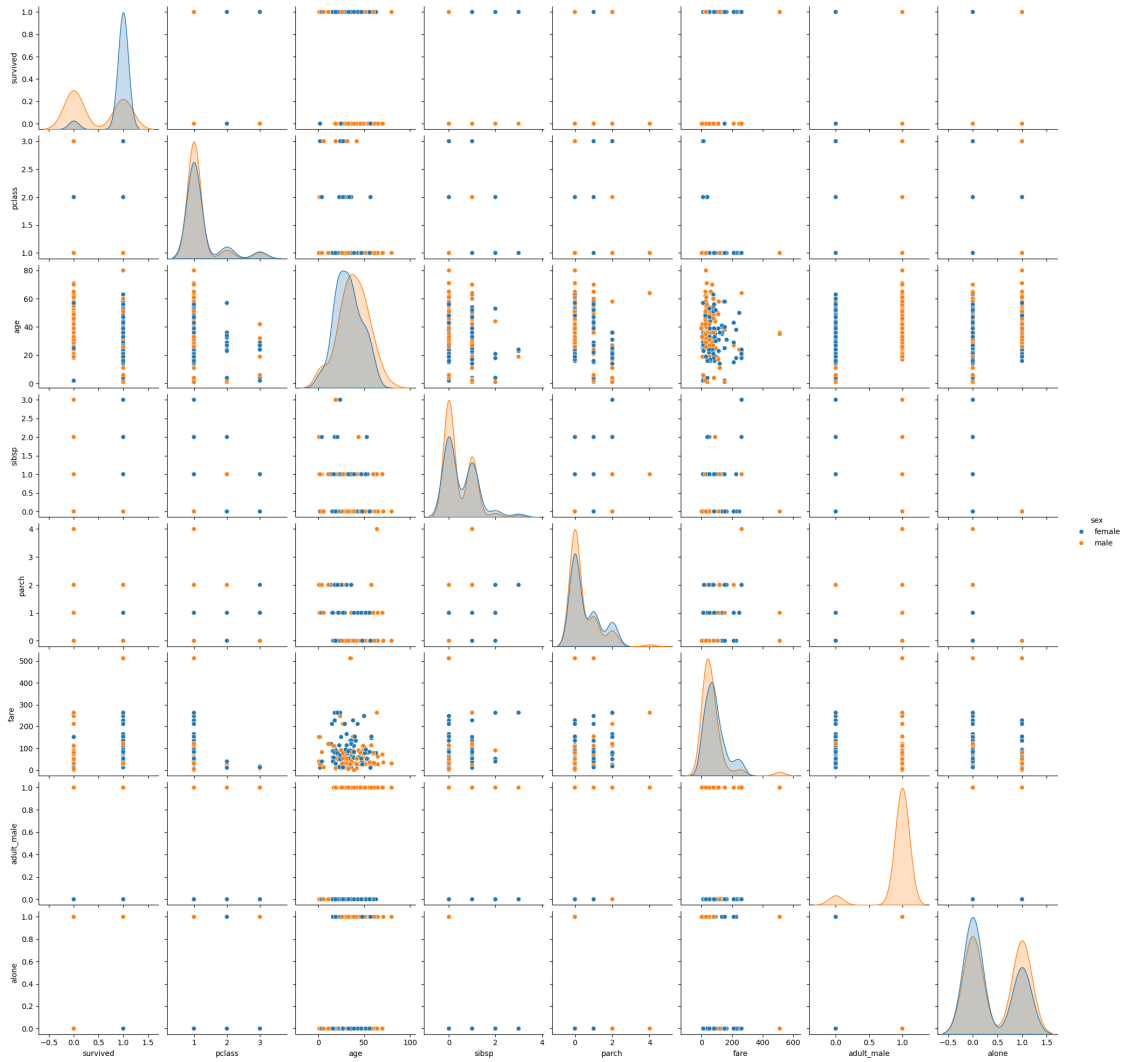


[ ]:
```
## Pair Plot for Dataset Features
```

[26]:
```
dataset = dataset.dropna()
sns.pairplot(dataset)
sns.pairplot(dataset, hue='sex')
```

[26]: <seaborn.axisgrid.PairGrid at 0x708b75a174a0>

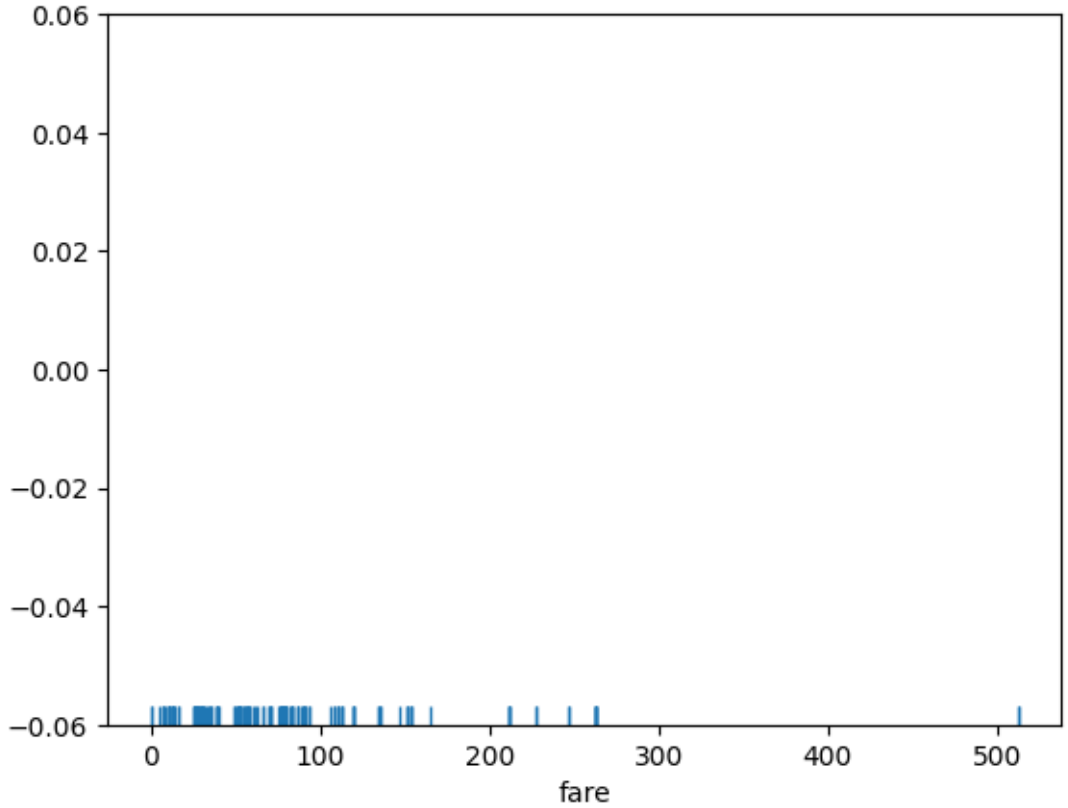

survived
pclass
age
sibsp
parch
fare
adult_male
alone



[7]:
```
## Rug Plot of Ticket Prices
```

[27]:
```
sns.rugplot(dataset['fare'])
```

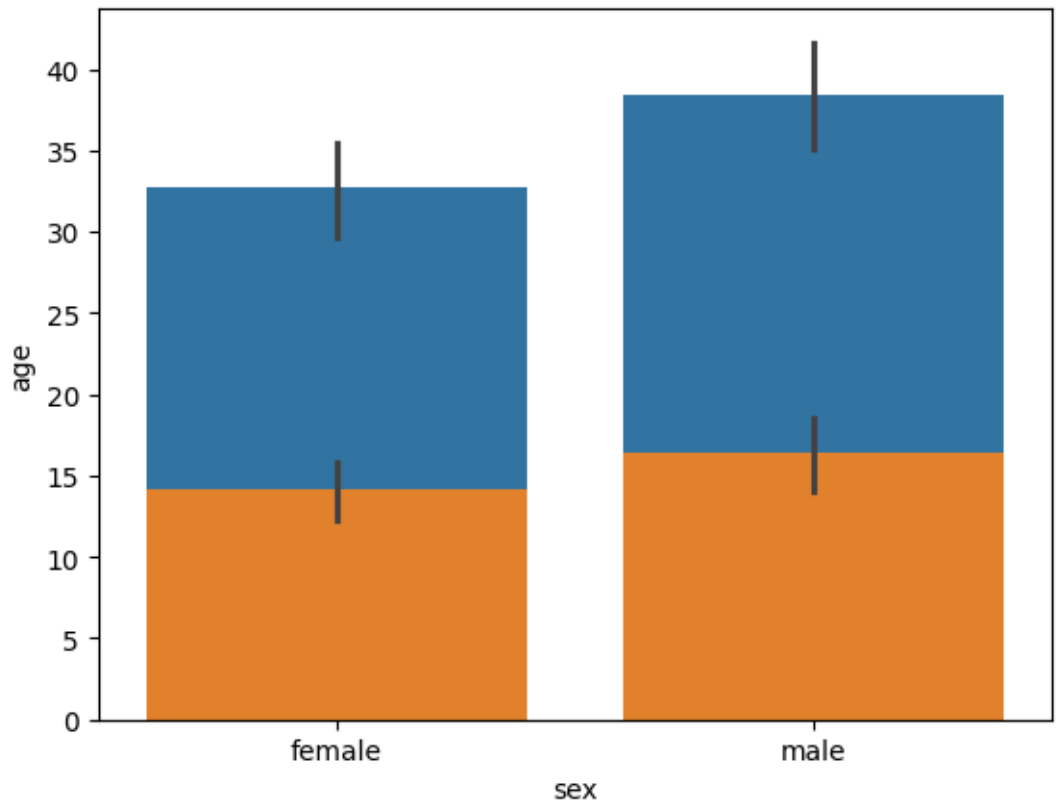
[27]: <Axes: xlabel='fare'>



[17]:
```
## Bar Plot of Average Age by Gender
```

[28]:
```
sns.barplot(x='sex', y='age', data=dataset)
sns.barplot(x='sex', y='age', data=dataset, estimator=np.std)
```

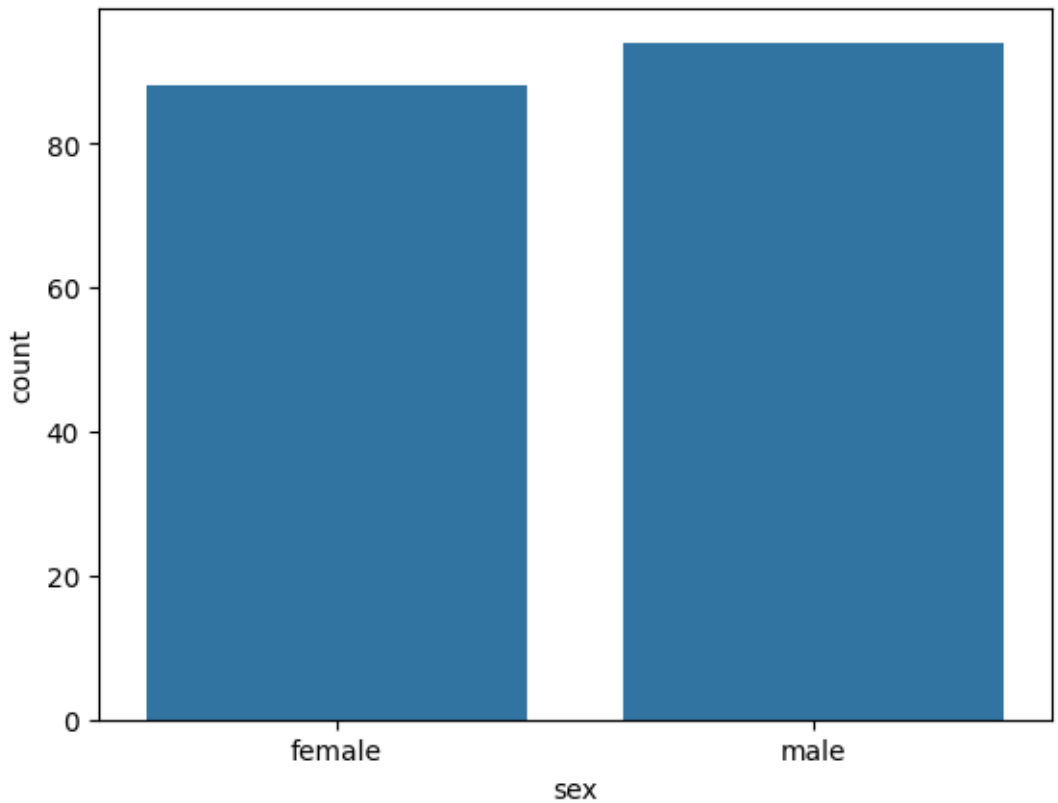
[28]: <Axes: xlabel='sex', ylabel='age'>



[18]:
```
## Count Plot of Passengers by Gender
```

[29]:
```
sns.countplot(x='sex', data=dataset)
```

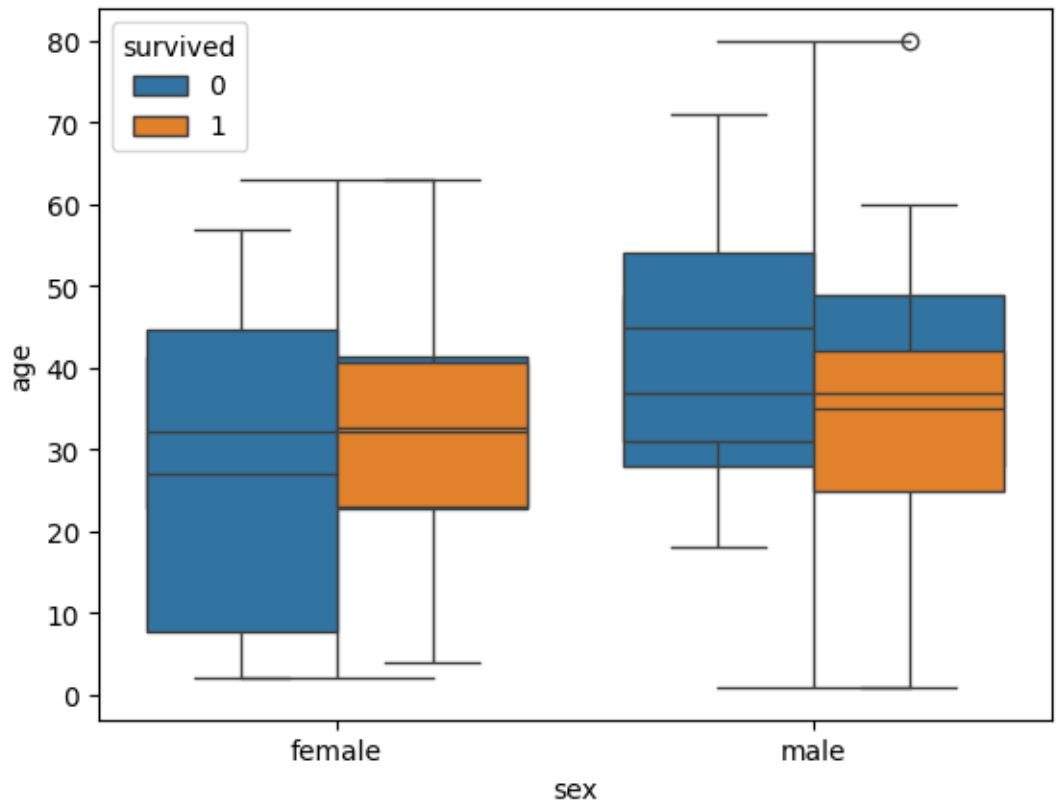
[29]: <Axes: xlabel='sex', ylabel='count'>



[30]:
```
## Box Plot of Age by Gender and Survival
```

[31]:
```
sns.boxplot(x='sex', y='age', data=dataset)
sns.boxplot(x='sex', y='age', data=dataset, hue="survived")
```

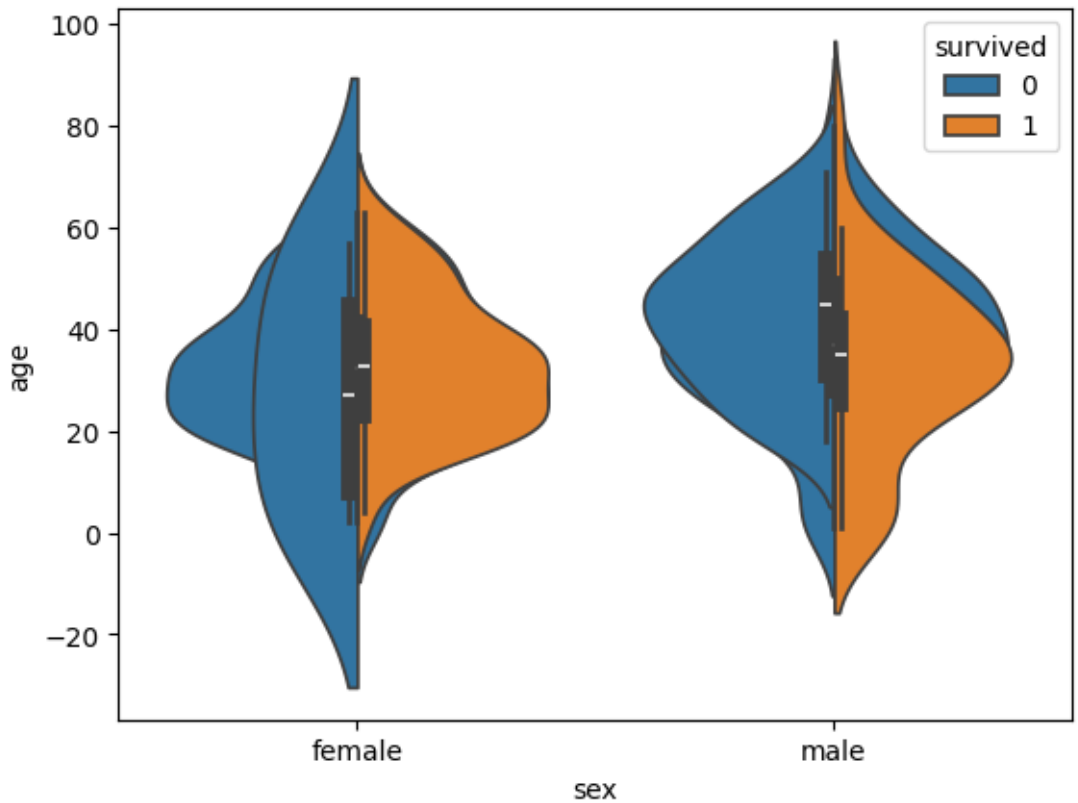
[31]: <Axes: xlabel='sex', ylabel='age'>



[32]:
```
## Violin Plot of Age Distribution by Gender and Survival
```

[33]:
```
sns.violinplot(x='sex', y='age', data=dataset)
sns.violinplot(x='sex', y='age', data=dataset, hue='survived', split=True)
```

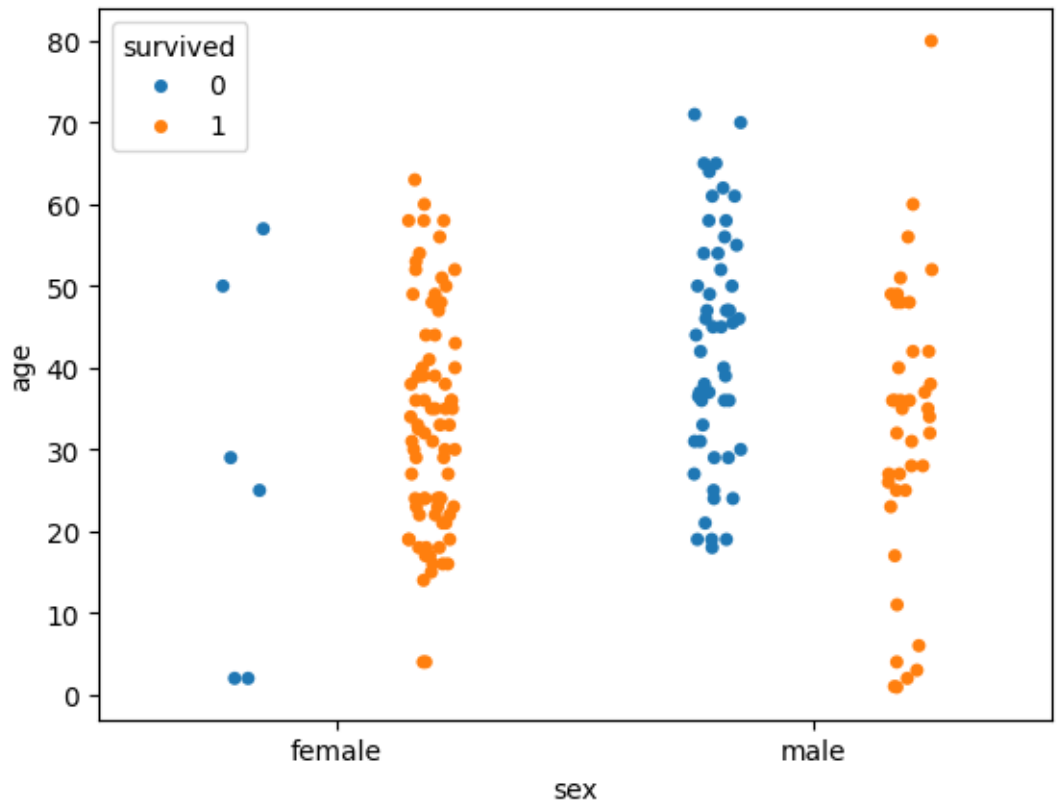
[33]: <Axes: xlabel='sex', ylabel='age'>



[34]:
```
## Strip Plot of Age by Gender and Survival
```

[41]:
```
sns.stripplot(x='sex', y='age', data=dataset, jitter=True, hue='survived',␣
↪dodge=True)
```

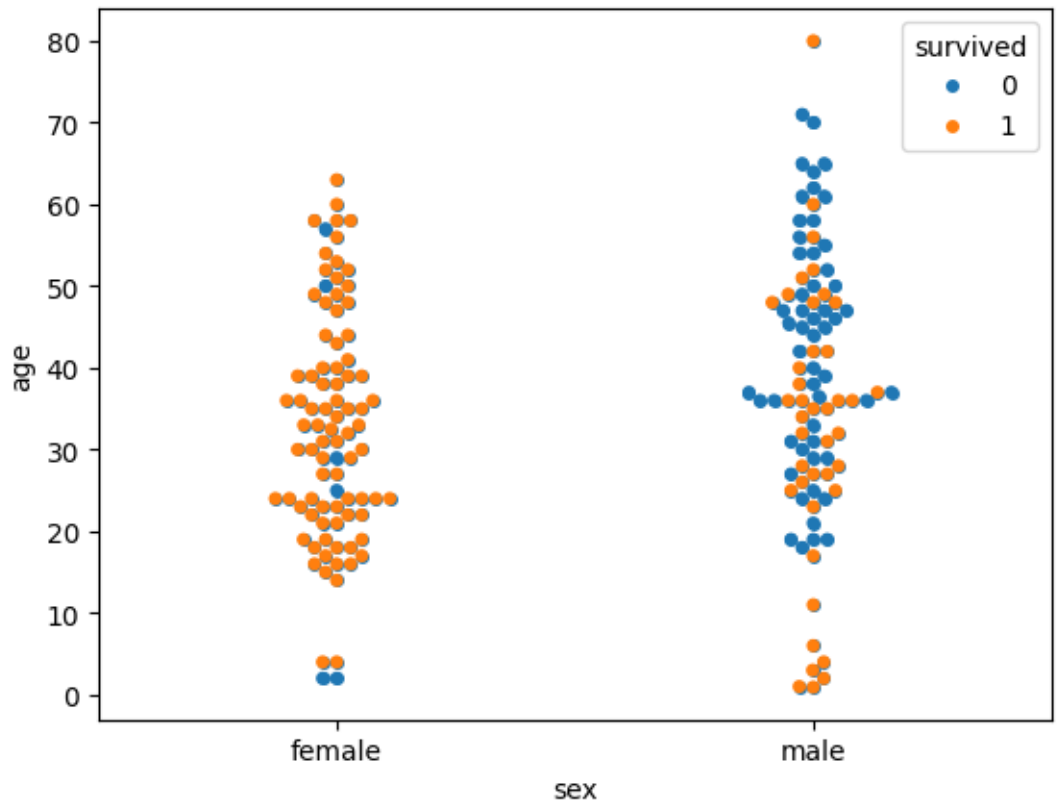
[41]: <Axes: xlabel='sex', ylabel='age'>



[ ]:
```
## Swarm Plot of Age by Gender and Survival
```

[37]:
```
sns.swarmplot(x='sex', y='age', data=dataset)
sns.swarmplot(x='sex', y='age', data=dataset, hue='survived')
```

[37]: <Axes: xlabel='sex', ylabel='age'>



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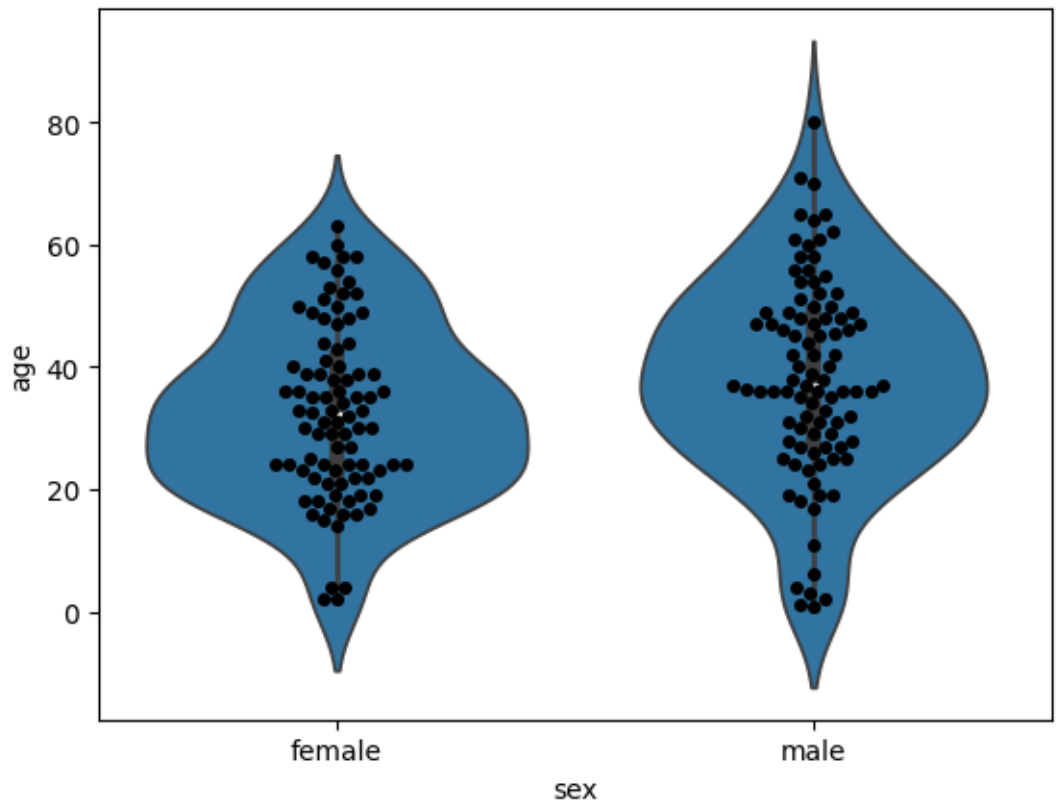
```
## Combined Swarm and Violin Plot
```

[38]:
```
sns.violinplot(x='sex', y='age', data=dataset)
sns.swarmplot(x='sex', y='age', data=dataset, color='black')
```

[38]: <Axes: xlabel='sex', ylabel='age'>


80
60
40
20
0
age
female
male
sex

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